

Stephen Matthew Griffies

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RESEARCH AND TEACHING ORCID GOOGLE SCHOLAR WEB OF SCIENCE

RESEARCH STATEMENT

I conduct research in theoretical ocean physics, with a focus on the ocean's role in the Earth's climate. I make use of theoretical concepts and mathematical methods, idealized process models, realistic numerical circulation models, and field measurements. Particular research topics include studies of Atlantic and Southern Ocean circulation dynamics; global and regional sea level patterns and changes; transport of matter and energy by mesoscale and submesoscale eddies; subgrid scale parameterizations of turbulent ocean stirring and mixing; analysis methods aimed at conceptually understanding ocean circulation and transport; physical and mathematical foundations of ocean circulation models; fundamentals of geophysical fluid mechanics.

EDUCATION STATEMENT

As a teacher, mentor, author, and journal editor, I aim to foster an understanding of physical concepts and their creative and rigorous use in describing observed and simulated ocean and climate phenomena. I am particularly interested in revealing how concepts and methods from theoretical physics can be leveraged to deepen understanding of earth climate mechanics, and for fostering an appreciation of geophysical fluid mechanics within the broader context of theoretical mechanics and thermodynamics. Teaching and mentoring are central to all of my work, and I strive to generate curiosity and passion in colleagues and students while fostering an honest and non-judgmental scientific pursuit of how nature works.

COLLABORATION STATEMENT

I nurture collaborations with scientists at all career stages who are passionate about furthering a rigorous understanding of ocean and climate physics, and with keen interests in theoretical ocean physics.

BROADER INTERESTS AND ACTIVITIES

meditation, yoga, walking, writing, sustainability, cultures, surfing, skiing

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EDUCATION

1995-1996	Postdoctoral fellow in ocean and climate physics (mentor: Kirk Bryan)	Princeton University
1993-1995	NOAA climate and global change fellow (mentor: Kirk Bryan)	Princeton University
1988-1993	PhD in theoretical high energy physics (mentor: Mirjam Cvetič)	University of Pennsylvania
1987-1988	pre-PhD studies in physics	University of Washington
1986-1987	Masters of science in engineering sciences & applied mathematics	Northwestern University
1981-1986	Bachelor of science in chemical engineering	Louisiana State University
1978-1981	High school (grades 10-12)	Biloxi High School, Mississippi

SPECIAL TOPIC SCHOOLS

Jan 1998	NATO Advanced Study Institute: Ocean Modeling and Parameterization	Les Houches, France
Feb 1995	NATO Advanced Study Institute: Climate Variability and Predictability	Les Houches, France
Jul 1994	Meeting of UCAR Global and Climate Change Fellows	Steamboat Springs, Colorado, USA
Jul 1992	Theoretical Advanced Study Institute: From String Theory to Black Holes	Boulder, Colorado, USA
Jul 1991	High Energy Physics and Cosmology School, Center for Theoretical Physics	Trieste, Italy
Jun 1991	Theoretical Physics Summer School: Particle Physics in the 1990's	Les Houches, France

EMPLOYMENT AND APPOINTMENTS

2026–present	CNRS Research Director, IPSL/LOCEAN (Choose France for Science Laureate), Paris, France
2026-present	Research Scholar, Princeton University's Atmospheric and Oceanic Sciences Program
2025	Lecturer with rank of professor, Princeton University's Atmospheric and Oceanic Sciences Program
2024–2025	Chair, Visiting Scientist Committee, NOAA/Princeton Cooperative Institute for Modeling the Earth System
2024–2025	Director of Graduate Studies for Princeton University's Atmospheric and Oceanic Sciences Program
2022–2024	Graduate Work Committee for Princeton University's Atmospheric and Oceanic Sciences Program
2020–2025	Team lead for the GFDL Ocean/Cryosphere Division's high resolution climate model project CM4X
2015–2026	Lecturer in Princeton University's Atmospheric and Oceanic Sciences Program
2013–2017	Member, GFDL Model Development Team Steering Committee
Jun-Aug 2012	Visiting Scientist, National Center for Atmospheric Research, Boulder, USA
Jan-Jun 2011	Distinguished Visiting Scientist Fellow, CSIRO Marine and Atmospheric Research, Hobart, Australia
Mar 2009	Visiting Professor, Universite catholique de Louvain, Belgium
Jan-Nov 2005	Visiting Scientist, CSIRO Marine and Atmospheric Research, Hobart, Australia
2001–2005	Leader of the GFDL Oceans and Climate Group
2000–2011	Co-lead of the GFDL Ocean Model and Climate Model Development Teams
1996-2025	Staff Physical Scientist, NOAA/GFDL (senior scientist 2011-2025, retired 1 May 2025)
1995–1996	Visiting Scientist, GFDL and Princeton University
1993–1995	NOAA Climate & Global Change Postdoc Fellow at Princeton University
1988–1993	Physics Graduate Research Fellow, University of Pennsylvania Physics Department
1986–1987	Engineering Sciences and Applied Mathematics Fellow, Northwestern University
1984–1986	Chemical Engineering Research Laboratory Technician, Louisiana State University

AWARDS AND HONORS

2026-2029	Choose France for Science Laureate awarded through CNRS at IPSL/LOCEAN, Paris
2022	NOAA Administrator's Award (with 26 others) "For advancing the understanding of the Earth System by developing and applying NOAA's state-of-the-art Coupled Carbon-Chemistry-Climate model"
2021	Reuters Hot List of Climate Scientists (#585)
2019	Department of Commerce Silver Medal Award (with Robert Hallberg and Matthew Harrison): "For developing the state-of-the-art Modular Ocean Model version 6 (MOM6) to strengthen the Nation's longer-range environmental prediction capabilities."
2019	Sigma Xi scientific honor society
various	Web of Sciences (Clarivate) Highly Cited Researcher (2018, 2020, 2021, 2022, 2023, 2024)
2017	Elected Fellow of the American Geophysical Union "For exceptional and sustained contributions to the understanding of large-scale ocean circulation and physics and seminal advances in ocean modeling"
2017	NOAA Administrator's Award (with Robert Hallberg) "For scientific leadership for the innovation of the versatile community-based Modular Ocean Model MOM6"
2014	European Geosciences Union Fridtjof Nansen Medal for Oceanographic Research "For outstanding contribution and leadership in ocean general circulation model development and critical insights in the physical nature and parameterization of ocean processes"
2013	Department of Commerce Silver Medal Award (with nine other GFDL staff scientists): "For development and application of NOAA's first comprehensive Earth System Model that couples the carbon cycle and climate for projection of changes"
2012	NOAA Administrator's Award "For scientific vision, leadership and development of the Modular Ocean Model (MOM4) for climate modeling, research and predictions"
2011	CSIRO Distinguished Visiting Scientist Fellow, Australia
2009	Visiting Professor, Universite catholique de Louvain, Belgium
2001	NOAA/Oceanic and Atmospheric Research Outstanding Scientific Review Paper
1999	NOAA/Oceanic and Atmospheric Research Outstanding Scientific Paper
1998	NOAA/Oceanic and Atmospheric Research Employee of the Year
1997	NOAA/Environmental Research Laboratories Outstanding Scientific Paper
1993-1995	NOAA Climate and Global Change Research Fellow

PROFESSIONAL SERVICES AND MEMBERSHIPS

2026–present	Co-lead of the NEMO Diagnostics Working Group
2024–present	Scientific advisory board for the C-Star program at [C]Worthy
2021-2025	NEMO Scientific Advisory Committee
2021-2026	Editor-in-Chief for AGU's Journal of Advances in Modeling Earth Systems (JAMES)
2018-2020	Ocean/Cryosphere Editor for AGU's Journal of Advances in Modeling the Earth System (JAMES)
2019-2023	Chair of the awards committee for the EGU Fridtjof Nansen Medal for Oceanographic Excellence
2017-2021	Advisory board for the TICTOC Project, United Kingdom
2016-2019	Awards committee for the EGU Fridtjof Nansen Medal for Oceanographic Excellence
2014-2018	WCRP/CLIVAR Scientific Steering Group
2014-2016	Co-lead of the NCEP Climate Model Development Task Force
2012-2014	CLIVAR/CliC/SCAR Southern Ocean Region Implementation Panel
2012-present	WCRP/CLIVAR Ocean Model Development Panel (ex officio)
2010-present	European Geosciences Union (EGU)
2009-2015	Scientific Advisory Board for the Catalan Climate Institute *IC3*, Barcelona, Spain
2007-2018	Editor of the journal **Ocean Modelling**
2006-2009	WCRP/CLIVAR Scientific Steering Group (ex officio)
2004-2009	WCRP/CLIVAR Working Group on Coupled Modelling (ex officio)
2004-2007	Editorial Board of the journal **Ocean Science**
1999-2012	WCRP/CLIVAR Working Group on Ocean Model Development (co-chair 2004-2009)
1993-present	American Geophysical Union and American Meteorological Society

UNIVERSITY TEACHING

- Autumn semester 2025: Princeton University Atmospheric and Oceanic Sciences 585: Special topics: **Methods in the Analysis of Ocean Scalar Properties**. 24 lectures of 80 minutes duration covering the full course, presenting a suite of theoretical tools to support physical ocean analysis. Topics include material tracers, tracer advection and diffusion, turbulent tracer transport, subgrid scale parameterizations, Green's function methods, watermass transformation analysis. Lecture material based on Griffies (2026): Geophysical Fluid Mechanics.
- Spring semester 2023, 2024, 2025: Princeton University Atmospheric and Oceanic Sciences 572: **Theory of Geophysical Fluid Waves and Instabilities**. 24 lectures of 80-90 minutes duration covering the full course, presenting theoretical foundations for ocean and atmosphere linear wave mechanics and instability theory. Topics include wave kinematics, acoustics, capillary waves, surface gravity waves, inertial waves, Rossby waves, shallow water waves, internal inertia-gravity waves, shear instability, inertial instability, symmetric instability, baroclinic instability. Lecture material based on Griffies (2026): Geophysical Fluid Mechanics.
- Autumn semester 2014 (0.5), 2015 (0.5), 2016 (0.5), 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024 (0.5): Princeton University Atmospheric and Oceanic Sciences 571: **Foundations of Geophysical Fluid Mechanics**. 24 (12) lectures of 80-90 minutes duration covering the full (half) of course, focusing on the theoretical foundations for geophysical fluid mechanics. Topics include mechanics of motion on a rotating planet, Eulerian and Lagrangian fluid kinematics, frictional stresses, pressure and form stress, buoyancy, Ekman mechanics, shallow water mechanics, vorticity and potential vorticity, quasi-geostrophy. Lecture material based on Griffies (2026): Geophysical Fluid Mechanics.
- Spring semester 2020: Princeton University Atmospheric and Oceanic Sciences 521: **Southern Ocean Seminar**. I provided five 90-minute lectures covering Southern Ocean dynamics, while other lecturers presented allied material to cover Southern Ocean science. Lecture material based on Griffies (2026): Geophysical Fluid Mechanics.
- Spring semester 2017, 2018, 2019, 2024: Princeton University Atmospheric and Oceanic Sciences 580: **Special Topics on Great Papers in Atmospheric and Oceanic Sciences**. I led on 90-minute discussion session on a chosen classic paper in ocean fluid mechanics.
- Spring semester 2016, 2019: Princeton University Geosciences 503: **Responsible Conduct of Research in Geosciences**. I co-taught one three-hour discussion session on ethical behavior in research.
- Autumn semester 1993: Princeton University Atmospheric and Oceanic Sciences Special Topics 580: **Data Assimilation in Atmospheric and Oceanic Models**. I was the co-lecturer and coordinator of visiting lectures throughout the semester.
- 1990–1993: Instructor, Undergraduate Physics Laboratory, University of Pennsylvania
- 1990–1993: Teaching Assistant, General Relativity and Quantum Field Theory, University of Pennsylvania

Mentoring, PhD committees, and sabbatical hosting

2027-2029	Jamie Hilditch	CNRS/LOCEAN/Paris postdoc
2027-2029	Oliver Street	CNRS/LOCEAN/Paris postdoc
2025-present	Hunter Camp	Princeton University graduate student (AOS committee member)
2024	Claire Yung	Visiting graduate student (from ANU, Canberra, AUS)
2024-present	Stefan Kildal-Brandt	Princeton University graduate student (geophysics committee member)
2023-present	Kiera Lowman	Princeton University graduate student (AOS committee member)
2023-present	Maxime Keutgen De Greef	Princeton University graduate student (AOS committee member)
2023-present	Kentaro Hanson	Princeton University graduate student (applied maths committee member)
2023	Jan Zika	Princeton University visiting scholar (from UNSW, Sydney, AUS)
2022-2027	Matthew Lobo	Princeton University graduate student (AOS, primary mentor)
2022-present	Winnie Chu	Princeton University graduate student (AOS committee member)
2022-2025	Wenda Zhang	Princeton University postdoc fellow
2021-2022	Rachel Pang	Princeton University undergraduate (junior paper mentor)
2021	Abigail Bodner	Brown University graduate student (PhD thesis reader)
2020-2025	Jan-Erik Tesdal	Princeton University postdoc fellow
2020	Ruth Moorman	Princeton University predoc intern
2019-2020	Benjamin Taylor	Princeton University predoc intern
2019-2021	Hemant Khatri	Princeton University postdoc fellow
2019-2020	Elizabeth Yankovsky	Princeton University graduate student (AOS committee member)
2019	Hussein Aluie	Princeton University visiting scholar (from University of Rochester)
2018-2022	Graeme MacGilchrist	Princeton University postdoc fellow
2017-2022	Houssam Yassin	Princeton University graduate student (AOS, primary mentor)
2017-2018	Laure Zanna	Princeton University visiting scholar (from Oxford University)
2017	Jianjun Yin	Princeton University visiting scholar (from University of Arizona)
2016-2019	Brandon Reichl	Princeton University postdoc fellow
2016-2018	Nathaniel Tarshish	Princeton University predoc intern
2015-2017	Amanda O'Rourke	University of Michigan postdoc fellow (with Brian Arbic, Un of Michigan)
2015-2016	Henri Drake	Princeton University predoc intern
2014-2017	Anna FitzMaurice	Princeton University PhD student (AOS committee member)
2014-2015	Ivy Frenger	Princeton University postdoc fellow (with Jorge Sarmiento)
2014	Magnus Hieronymus	Stockholm University graduate student (PhD opponent)
2013-2017	Robert Nazarian	Princeton University PhD student (AOS committee member)
2013-2016	Adele Morrison	Princeton University postdoc fellow (with Jorge Sarmiento)
2013	Terrence O'Kane	GFDL visiting scholar from CSIRO Marine Research, Hobart, Australia
2012-2017	Carolina Dufour	Princeton University postdoc fellow (with Jorge Sarmiento)
2012-2013	Yalin Fan	Princeton University postdoc fellow
2011-2014	Michael Bueti	University of Rhode Island PhD student (committee member with Isaac Ginis)
2008-2011	Michael Bates	University of New South Wales PhD student (committee member with M. England)
2005-2009	Andreas Klocker	University of Tasmania PhD student (committee member with T. McDougall)
2003-2004	Rüdiger Gerdes	GFDL visiting scholar (from AWI, Bremerhaven, Germany)
2001-2002	Harper Simmons	Princeton University postdoc fellow
1999-2002	Shafer Smith	Princeton University postdoc fellow

Oceanographic field work

- Mar-May 2017: Eight week cruise on the *RRS JC Ross* to the Orkney Passage and Scotia Sea, as part of the Dynamics of the Orkney Passage Outflow (DynOPO) project. Principal Scientific Officer: Alberto Naveira Garabato.
- Jul 1993: Two week cruise on the *CCGS Hudson* to the Labrador Sea in support of the WOCE Line AR7W Atlantic Circulation Experiment. Chief Scientist: John Lazier.

Participation in research grants (USA, Australia, and France)

- 2026-2032: Partner Investigator, Australian Research Council Centre of Excellence in Our Future Oceans, AU$44M.
- 2026-2029: Laureate, Choose France for Science, Agence Nationale de la Recherche + CNRS, €2.4M.
- 2021-2025: Partner Investigator, Australian Research Council Centre of Excellence in Antarctic Science, AU$25M.
- 2024-2025: PI, NOAA/NSF Climate Process Team: *Ocean Transport and Eddy Energy*, sub-award with NYU, $150K.
- 2020-2024: Co-PI, NOAA/NSF Climate Process Team: *Ocean Transport & Eddy Energy*, $770K.
- 2020-2023: Co-PI, NOAA: *Drivers of coastal sea level change along the eastern US*, $200K.

- 2020-2022: PI, US DOE: *Diagnostics and Performance Metrics for Evaluating Ventilation Pathways and Interior Water Mass Properties in Ocean Models*, $180K.
- 2019-2022: Co-PI, Australian Research Council: *Risks of rapid ocean warming at the Antarctic continental margin*, AU$582K.
- 2018-2022: Co-PI, NOAA: *Addressing Key Issues in CMIP6-era Earth System Models*, $434K.
- 2017-2023: Partner Investigator, Australian Research Council Centre of Excellence for Climate Extremes, AU$30M.
- 2016-2018: Co-PI, NOAA: *Development toward NCEP's fully-coupled global forecast and data assimilation system: A coupled wave-ocean system*, $316K.
- 2016-2020: Partner-Investigator, Australian Research Council: *An Australian Consortium for Eddy-Resolving Ocean-Sea Ice Modelling*, AU$599K.
- 2014-2020: Partner-Investigator, US NSF: *Southern Ocean Carbon & Climate Observations & Modeling (SOCCOM)*, $20.9M
- 2014-2017: Co-PI, US DOE: *Three-dimensional structure of the Southern Ocean overturning circulation*, $624K.
- 2014-2017: Co-PI, NASA: *The role of mesoscale eddies in cross-frontal transport and subduction of nutrients and carbon in the Southern Ocean*, $715K.
- 2013-2016: Co-PI, NOAA: *Signature of the Atlantic meridional overturning circulation in the North Atlantic dynamic sea level*, $393K.
- 2011-2015: Co-PI, US DOE: *Mode and intermediate waters in Earth System Models*, $519K.
- 2011-2018: Partner Investigator, Australian Research Council Centre of Excellence for Climate System Science, AU$21.4M.
- 2010-2015: Co-PI, NOAA and US NSF: *Climate Processes Team on representing internal-wave driven mixing in global ocean models*.
- 2003-2008: Co-PI, NOAA and US NSF: *Climate Processes Team on ocean eddy mixed layer interactions*.
- 2003-2008: Co-PI, NOAA and US NSF: *Climate Processes Team on gravity current entrainment*.

Research presentations since 2010

- Oct 2026: Problems and prospects at the frontier of ocean climate model theory and practice, Invited talk at IOW, Warnemünde, Germany.
- Sept 2026: Problems and prospects at the frontier of ocean climate model theory and practice, Virtual seminar to Imperial College, London.
- Jun 2026: Climate model thermal equilibration in an ocean mesoscale-dominant regime, seminar to LOCEAN, Paris.
- Mar 2025: Climate model thermal equilibration in an ocean mesoscale dominant regime, virtual talk to the Harvard University Earth and Planetary Sciences Department.
- Dec 2024: MOM6 and the CM4X Climate Model Hierarchy, talk given at the annual meeting of the American Geophysical Union, Washington, DC, USA.
- Oct 2024: MOM6 and the CM4X Climate Model Hierarchy, talk given at the Center for Ocean/Atmosphere Science, Courant Institute of Mathematical Sciences, New York University, USA.
- Sep 2024: MOM6 and the CM4X Climate Model Hierarchy, talk given at the COMMODORE workshop on numerical methods in ocean modeling, National Center of Atmospheric Research, Boulder, USA.
- Aug 2024: MOM6 and the CM4X Climate Model Hierarchy, talk given at the annual Eddy Energy and Transport Climate Process Team meeting, Brown University, USA.
- June 2024: Modeling the ocean with MOM6: A primer on vertical Lagrangian remapping and the analysis of finite volume ocean models, virtual seminar given at Alfred Wegener Institute, Germany, in honor of the retirement of Rüdiger Gerdes.
- April 2024: Modeling the ocean with MOM6: A primer on vertical Lagrangian remapping and the analysis of finite volume ocean models, virtual seminar given at University of Rhode Island Graduate School of Oceanography.
- Feb 2022: A mathematical formalism for circulation in water mass configuration space, virtual presentation at AGU Ocean Sciences Meeting.
- May 2020: The importance of refined model resolution for understanding and projecting global, regional, and coastal sea level, NASA GISS virtual Sea Level Seminar Series.
- Jan-April 2020: Vertical Lagrangian-remapping, generalized vertical coordinates, and spurious diapycnal mixing in ocean models: COMMODORE meeting, Hamburg, Germany; DRAKKAR meeting, Grenoble, France; AGU Ocean Sciences, San Diego, USA; CESM Ocean Model Working Group.
- Oct 2019: Water mass transformation (WMT) analysis and tracer budgets with generalized vertical coordinates and vertical Lagrangian-remapping, annual meeting of the Transient tracer-based Investigation of Circulation and Thermal Ocean Change (TICTOC) Project, Exeter, UK.
- Feb 2019: water mass transformation analysis in ocean models: some thoughts and questions, workshop on Water mass transformation for ocean physics and biogeochemistry, University of New South Wales, Sydney, Australia.
- May 2018: Understanding and projecting global, regional, and coastal sea level: Reasons to include coastal ocean processes in global models: Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA) Annual Meeting, Australian

National University, Canberra, Australia & University of New South Wales, Sydney, Australia; ISSI workshop on understanding the relationship between coastal sea level and large-scale ocean circulation, Bern, Switzerland.

- Feb 2018: Subsurface Warming of Antarctic Coastal Waters: a Role for Both Winds and Freshening: American Geophysical Union Ocean Sciences Conference, Portland, Oregon, USA.
- Dec 2017: Localized Rapid Warming of West Antarctic Subsurface Waters by Remote Winds: American Geophysical Union Fall Meeting, New Orleans, Louisiana, USA.
- Nov 2017: Physical mechanisms of sea level variations in a changing climate: International CLIVAR Scientific Steering Group meeting, Indian Institute of Tropical Meteorology, Pune, India.
- Jul 2017: Localized rapid warming of West Antarctic subsurface waters by remote winds: WCRP Conference on Regional Sea-level Changes and Coastal Impacts, Columbia University, New York City, USA.
- May 2017: Localized rapid warming of West Antarctic subsurface waters by remote winds: *RRS JC Ross* research cruise JR16005 to Orkney Passage, Southern Ocean.
- Jan 2017: The ocean mesoscale: observations, theory, and modeling: Banff International Research Station (BIRS) workshop: *Transport in unsteady flows: From deterministic structures to stochastic models and back again*, Banff, Canada.
- July 2016: Elements of sea level in a changing climate: Indian Institute of Tropical Meteorology, Pune, India.
- May 2016: Elements of sea level in a changing climate: University of New South Wales, Sydney, Australia & Australian National University, Canberra, Australia.
- Jan 2016: Elements of sea level in a changing climate: Louisiana State University Chemical Engineering Department, Baton Rouge, Louisiana, USA.
- Oct 2015: Impacts on ocean heat from the mesoscale: Lamont-Doherty Earth Observatory / Columbia University, USA.
- Oct 2015: Impacts on ocean heat from the mesoscale: Stony Brook Marine Sciences, Stony Brook, USA.
- Oct 2014: Impacts on ocean heat from the mesoscale: Meeting on ocean heat uptake at the National Oceanography Centre, Southampton, UK.
- Jun 2014: Impacts on ocean heat from the mesoscale: University of Stockholm, Sweden.
- Feb 2013: Sea level in a suite of forced global ocean-ice simulations: CLIVAR workshop on Sea-Level Rise, Ocean/Ice-Shelf Interactions, and Ice Sheets, Hobart, Australia
- Jan 2013: Ocean model numerics and physics: challenges for mesoscale eddying global climate simulations: 10th annual meeting of the Drakkar Ocean Modelling Consortia, Grenoble, France
- Sep 2012: Sea level in ocean climate models: fundamentals and practices: University of Tasmania, Hobart, Australia
- Mar 2011: Dynamic sea level, static sea level, and the non-Boussinesq steric effect: Australia National University, Canberra, Australia
- Sep 2010: Sensitivity of Atlantic ocean variability to ocean physics and vertical coordinate: CLIVAR WGOMD/GSOP Workshop on Decadal Variability, Predictability, and Predictions: Understanding the Role of the Ocean. Boulder USA

INVITED PEDAGOGICAL LECTURES AND SPECIAL TOPICS COURSES

- Oct 2026: The ocean at the heart of a warming climate. General audience lecture given at LOCEAN, Paris, France.
- Jun 2026: Fundamentals of ocean models and the analysis of ocean simulations. Five lectures (two hours each) concerning the ocean equations forming the foundations of NEMO. Paris, France.
- May 2026: A primer on the ocean mesoscale and its parameterization in circulation models: Sampling the physics/maths/numerics (with a focus on tracers), talk given at the Institut Henri Poincaré workshop: *Instabilities and Transitions in Geophysical Flows*.
- Sep 2024: The Long and Winding Road to OMIP, talk given at the 50th anniversary of the Ocean Section of NCAR, National Center of Atmospheric Research, Boulder, USA.
- Oct 2022: Fundamental equations and diagnostics for MOM6. Lecture given as part of a 2-day tutorial on the Modular Ocean Model (MOM6). Princeton, USA.
- Feb 2022: meditation for scientists (with Jonathan Lilly): American Geophysical Union Ocean Sciences Conference. Honolulu, Hawaii (virtual).
- Feb 2020: meditation for scientists (with Jonathan Lilly): American Geophysical Union Ocean Sciences Conference, San Diego, CA.
- April/May 2019: Fundamentals of ocean models and the analysis of ocean simulations. 15 lectures (45 minutes each) on ocean model fundamentals and analysis methods given as part of the **Advanced Ocean Modelling Summer School**. Tasmania, Australia.
- Jan 2019: A historical survey of neutral diffusion methods and comments on current research, presented during the celebration of Peter Gent's career, NCAR, Colorado, USA.
- Jan 2019: Ocean circulation as a problem in mathematical & computational physics: a historical and contemporary perspective. Public lecture given as part of the Australian Mathematics Science Institute (AMSI) Summer School at the University of New South Wales, Sydney, Australia.

- July 2016: Ocean modelling: an introduction for mathematical physicists: Department of Mathematics, Savitribai Phule Pune University, Pune, India.
- Jul 2016: Ocean Modelling and sea level analysis: three lectures (two hours each) at the International Centre for Theoretical Physics / Indian Institute for Tropical Meteorology: Advanced School on Earth System Modelling, Pune, India
- Apr 2014: Problems and prospects with ocean mesoscale eddying climate models: Nansen Medal lecture at the European Geosciences Union annual meeting, Vienna, Austria.
- Apr 2014: Problems and prospects with ocean mesoscale eddying climate models: lecture given at a CLIVAR workshop on eddying ocean climate models, Kiel, Germany.
- Sep 2013: Problems and prospects of model comparisons: an ocean process perspective: lecture given at a symposium celebrating the 80th birthday of Gerold Siedler, Kiel, Germany.
- Aug 2013: Ocean models and ocean modeling: lectures on the fundamentals and practices: Five lectures (two hours each) at the International Centre for Theoretical Physics School: Fundamentals of Ocean Climate Modeling at Global and Regional Scales, Hyderabad, India
- Sep 2012: Ocean modelling with MOM and its relation to Australian Ocean Climate Science: Second meeting of Consortia for Ocean Modelling in Australia, Hobart, Australia
- Feb 2012: Ocean modelling with MOM and its relation to Australian Ocean Climate Science: First meeting of Consortia for Ocean Modelling in Australia, Hobart, Australia
- Nov 2010: Ocean Climate Modeling at GFDL: Scientific Workshop for the Centre for Australian Weather and Climate Research, Hobart, Australia
- Mar 2009: Physical Processes Setting the Ocean's Water Masses: four lectures (two hours each) at the Université Catholique de Louvain, Belgium
- Nov 2007: Ocean Model Fundamentals: 10 lectures (two hours each) at the University of Tasmania, Australia
- Aug 2006: Ocean Model Fundamentals: two lectures (one hour each) at the NSF summer school, Modern Mathematical Methods in Physical Oceanography, Breckenridge, USA
- Oct 2004: Ocean Model Fundamentals: 10 lectures (two hours each) at the Indian Intensive School on Large-Scale Ocean Modelling, Bangalore, India
- Sep 2004: Ocean Model Fundamentals: three lectures (two hours each) at the Global Ocean Data Assimilation Experiment Summer School, La Londe Les Maures, France
- May 2003: Ocean Climate Modeling at NOAA-GFDL: two lectures (one hour each) for a workshop on ocean modeling, Hobart, Australia
- May 2002: Ocean Climate Modeling with MOM4: three lectures (one hour each) for a workshop on ocean modeling, Kiel, Germany
- Jan 2001: Ocean Dynamics and Modeling: three lectures (two hours each) at La Escuela de Verano de Universidad de Concepción, Chile
- Mar 1999: Ocean and Climate Modeling: two lectures (90 minutes each) at Conference on Global Climate, Barcelona, Spain

DOCUMENTS UNDER REVIEW OR IN PREPARATION

7. Geophysical Fluid Mechanics, 2028: S.M. Griffies, *multi-volume book project in preparation*, with anticipated completion in 2028.
6. Regional estimates of oceanic excess temperature change, 2026: E. McDonagh, A. J. G. Nurser, Q. Wu, J. Gregory, S. M. Griffies, T. W. N. Haine, J. Church, L. Clément, H. Graven, H. Hewitt, Y. Kostov, A. Marzocchi, M.-J. Messias, E. Newsom, T. Sohail, C. Turner, L. Zanna, J. Zika, *in prep for Journal of Climate*
5. Numerical Mixing in Finite Volume Ocean Models Part 1: Derivation and implementation of diagnostic formula, 2026: J. D. Zika, J. I. Bisits, H. F. Drake, R. M. Holmes, S. M. Griffies, A. H. Gibson, A. McG. Hogg, J.-E. Tesdal, *in prep for Journal of Advances in Modeling Earth Systems (JAMES)*.
4. Model sensitivity to the vertical distribution and latent heat loss from Antarctic meltwater, 2026: P. Colombo, M. H. England, C. K. Yung, H. R. S. Dawson, F. Boeira Dias, A. K. Morrison, A. E. Kiss, P. Spence, A. McC. Hogg, B. K. Galton-Fenzi, and S. M. Griffies, *submitted to JAMES*, https://essopenarchive.org/doi/full/10.22541/essoar.15007361/v1
3. Time averaging for tendency budget analysis, 2026: C. Bladwell, R.M. Holmes, J.D. Zika, A. E. Kiss, and S.M. Griffies, *submitted to Geoscientific Model Development*. https://editor.copernicus.org/EGUsphere/ms_records/egusphere-2026-3672
2. Steric sea-level response to Antarctic freshwater forcing, 2026: J. Milward, R. L. Beadling, J. P. Krasting, J. Steinberg, S. M. Griffies, J.-E. Tesdal, G. M. MacGilchrist, T. Martin, and J. MacIsaac, *in revision at Journal of Geophysical Research: Oceans*
1. The contribution of sterodynamic changes to the uncertainty of regional sea-level projections over the 21st century, 2026: J.-E. Tesdal, J. P. Krasting, R. E. Kopp, T. H. J. Hermans, P. Kumar, W. V. Sweet, S. M. Griffies, *in revision at AGU Advances*, https://doi.org/10.22541/essoar.174

PEER-REVIEWED PUBLICATIONS

2026

181. Ocean energy pathways: a multi-scale perspective, 2026: H. Aluie, R. Barkan, S.M. Griffies, D. Marshall, B. Storer, *Annual Review of Fluid Mechanics*, **59**, doi:10.1146/annurev-fluid-112723-054411

180. Steric-setup effects on US East Coast sea-level changes, 2026: J. Steinberg, S. M. Griffies, and C. Piecuch, *Geophysical Research Letters*, doi:10.1029/2026GL124322

179. Forced-dissipative two-layer quasi-geostrophic waves, instability, and turbulence over an aligned sloping bottom, 2026: M. Lobo and S. M. Griffies, *Journal of Physical Oceanography*, **53**, doi:10.1175/JPO-D-25-0205.1

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